

BOSS — Business Opportunity Screening System

Expert Knowledge Acquisition SRS

Status: Knowledge acquisition completed and the prototype design finalized for implementation as a simplified Prolog expert system with 25 fixed facts, 25 expert rules, lecture-aligned certainty factors, seven short interaction screens, an explanation facility, and six validation cases.

1. Expert Profile and Domain Experience

Expert full name: *«expert name»*

Current role/occupation: Independent small-business advisor and consultant; part-time mentor on an SME development programme

Organisation/business role(s): Owner-operator of a small services firm (2015–2021); advisor/mentor to owner-managed businesses; occasional minority investor in two small ventures

Years of business/management experience: 14

Countries/markets in which you have operated: Sri Lanka (Colombo, Gampaha, Kandy, Galle districts); limited remote/online work with clients in India and the Middle East

Types of businesses you have started, managed, advised, or invested in:

Tuition and skills-training centres; salons and personal-care services; small food outlets and cloud kitchens; grocery and hardware retail; online resellers and small e-commerce sellers; freelance/agency-style professional services (design, accounting, IT support); two small event-services businesses.

Which areas best describe your practical expertise?

- ☒ Starting new businesses
- ☒ Evaluating business ideas
- ☒ Service businesses
- ☒ Education/tuition businesses
- ☒ Pricing
- ☒ Marketing/customer acquisition
- ☒ Financial planning/cost control
- ☒ Staffing and management
- ☒ Expansion/branch decisions
- ☒ Location selection

- ☒ Risk management
- ☒ Other: Working-capital and cash-cycle planning for micro and small businesses

In one or two sentences, how would you describe your approach to deciding whether a new business opportunity is worth pursuing?

I first look for evidence that somebody will actually pay — not that the idea is good, but that money has changed hands or is about to. After that I check whether the owner can survive long enough to reach that money: enough capital, enough personal runway, and enough of the right experience. If both hold, everything else is a matter of degree.

2. B0SS Purpose, Users, and Scope

Expert confirmation

Is the above purpose appropriate? ☐ Yes ☐ No ☒ Modify

Modified purpose in my own words:

B0SS should give a structured first-pass opinion on whether a proposed small business is worth investigating further, and — more importantly — should tell the person exactly which assumptions they have not yet tested. Its main value is not the verdict but the list of things they still need to find out. It screens; it does not decide.

Who should use B0SS?

- ☒ First-time entrepreneur
- ☒ Existing small-business owner
- ☒ Manager considering a new venture
- ☐ Investor/partner performing an initial screen
- ☒ Student/learner
- ☒ Other: Business advisors/mentors, as a structured checklist during a first client meeting

Note: excluded investors because an investor screen needs valuation, ownership and exit questions that this knowledge base does not contain.

Which business types should B0SS evaluate?

- ☒ Small service businesses
- ☒ Small retail businesses
- ☒ Online/digital businesses

- ☒ Education/training businesses
- ☒ Professional services
- ☒ Other: Small food and beverage outlets, cloud kitchens, and owner-operated trades

Which business types should B0SS NOT evaluate?

Anything where the decision depends on specialist regulatory or technical judgement my rules do not contain: pharmaceuticals and medical practice, financial services and lending, insurance, alcohol and tobacco, firearms, large-scale import/export, manufacturing needing significant plant, agriculture and aquaculture with biological/seasonal risk, construction contracting, cryptocurrency and forex trading, and any franchise where a franchise agreement dictates the economics. Also exclude businesses with more than about 20 employees or capital above roughly LKR 25 million — that is beyond “small” and beyond my rules.

At what stage should B0SS be used?

- ☒ Idea stage
- ☒ Before committing capital
- ☐ Before launching
- ☒ Early-stage review
- ☒ Other: Before a second location/branch decision by an existing owner

Note: “before launching” is too late — by then money is usually already spent and the screening question has become a rescue question.

What should B0SS never claim or decide?

It must never predict revenue, profit or success probability as a number. It must never say a business “will work” or “will fail”. It must never give legal, tax, accounting or licensing advice, or tell someone whether to take a loan, pledge property, or use retirement savings. It must never advise on hiring, firing or partner disputes involving named people. And it must never present its recommendation as a substitute for talking to real customers.

3. Expert’s Natural Business-Opportunity Evaluation Process

3.1 First questions

1. Who has already paid you, or told you they would pay you, for this — and how much did they say?
2. How much money do you need in total to open and to run for six months before it supports itself, and how much of that do you actually have in hand right now?

3. What will you and your family live on for the next six to twelve months if this business earns nothing?
4. Who is already doing this near you or online, what do they charge, and why would a customer leave them for you?
5. Have you done this work yourself before — not managed it, done it?

3.2 Evaluation order

1. Demand and evidence of willingness to pay
2. Owner's capability, experience and available time
3. Money: capital required vs capital available, including six months of working capital
4. Personal survival runway of the owner and household
5. Competition and differentiation
6. Operations: location or channel, suppliers, staff, licensing

Why does this order matter?

Because the earlier items can end the conversation on their own and the later ones cannot. If nobody wants to buy, no amount of good location fixes it. If the owner cannot do or supervise the work, a good market just means someone else will take it. Money comes third because capital problems are sometimes solvable by shrinking the plan, whereas absent demand is not. I check the personal runway before competition because I have seen more small businesses die from the owner running out of household money than from a competitor.

3.3 Critical factors

#	Factor	Why
1	Evidence of paying demand	It is the only factor with no substitute. Everything else can be improved; a market that does not exist cannot be improved.
2	Capital adequacy including six months' working capital	Most failures I have seen were correctly-conceived businesses that ran out of cash before demand matured. Startup cost alone is a misleading number.
3	Owner's direct hands-on experience in the same work	In a small business the owner is the quality control, the sales team and the cost controller. Inexperience shows up as waste and as promises to customers that cannot be kept.
4	Owner's personal financial runway	An owner under household pressure takes short-term decisions — discounting, borrowing expensively,

#	Factor	Why
		cutting quality — that damage the business permanently.
5	Gross margin and cash cycle	A business can be profitable on paper and still fail if customers pay in 60 days and suppliers demand cash. Margin decides whether mistakes are survivable.

3.4 Immediate red flags / deal-breakers

1. The business cannot legally operate without a licence or approval the owner cannot realistically obtain.
2. Startup and six-month working capital is funded by high-interest informal borrowing, pawned jewellery, or a loan against the family home.
3. There is no evidence of demand at all and the owner is unwilling to test it before committing money.
4. The owner has neither done the work nor secured someone who has, in a business where the work is the product.
5. The whole business depends on one customer, one supplier, or one platform account that can be withdrawn without notice.
6. The owner's household has less than three months of living expenses and the business cannot pay them within that time.
7. Expected gross margin is below the cost of the money being used to fund the business.

Are any of these absolute deal-breakers even when other factors are strong?

Yes — items 1, 2 and 7 are absolute. A legal blocker is binary. Expensive informal debt converts a slow start into a personal catastrophe, and I will not endorse it however good the idea is. And if the margin is below the funding cost, the business mathematically destroys money as it grows, so strength elsewhere makes it worse, not better. Items 3, 4, 5 and 6 are strong negatives but can be repaired before launch, so they should produce NOT RECOMMENDED *at present* rather than a permanent no.

3.5 Must-have conditions

1. At least one concrete piece of evidence that a real customer will pay the intended price — a deposit, a signed enquiry, a waiting list, a previous customer, or a competitor visibly turning work away.
2. Capital in hand or firmly committed covering full startup cost plus six months of operating cost, from a source that does not endanger the household.

3. The owner can do the core work, or has a named, committed person who can, and can judge whether it is done well.
4. Gross margin high enough to absorb a bad first quarter — as a rule of thumb, above 30% for retail and above 45% for services.
5. The owner can give the business the hours the business actually needs, not the hours they have spare.

3.6 Compensating factors and exceptions

Example 1: Weakness strong competition can be compensated by a specific, defensible differentiation and an existing customer base the owner already serves. Reason: In small local markets, competition is rarely about being better overall; it is about owning one narrow reason to be chosen — a location, a skill, a language, a delivery time, a relationship. If the owner already has customers who follow them, the competitor count matters much less than the count suggests.

Example 2: Weakness owner's limited management experience can be compensated by very low fixed costs and a small, reversible starting scale. Reason: Inexperience is mostly a cost of learning. If fixed costs are low and the owner can stop within a month without large loss, they can afford to learn on the job. The same inexperience with a three-year lease and five employees is fatal.

Example 3: Weakness low capital availability can be compensated by an advance-payment or pre-order cash cycle and a home-based or shared-premises start. Reason: Capital requirement is not fixed; it is a consequence of the model chosen. A tuition class collecting fees in advance at home needs almost nothing. The same subject taught in a rented hall with staff needs a great deal. When capital is short, I ask whether the model can be changed before I say no.

Example 4: Weakness no prior business ownership experience can be compensated by deep hands-on trade experience plus a mentor or experienced partner for the commercial side. Reason: The technical and commercial halves are separable. A skilled baker with an accountant sibling is a very different risk from a skilled baker alone.

What common “rules of thumb” have important exceptions?

- *“Never enter a crowded market.”* Wrong when the market is crowded with poor operators and the owner is visibly better, or when demand is growing faster than the number of competitors. Many crowded markets are crowded because they work.
- *“Location is everything.”* Only true for walk-in retail and food. For services, tuition, and online businesses, reputation and channel access matter far more, and paying for a prime location is often the mistake itself.
- *“Start small and grow.”* Wrong where the business has a minimum viable scale — a restaurant kitchen or a gym cannot be half-built. Starting too small there produces a bad customer experience that permanently damages the reputation.

- *“Passion carries you through.”* Passion sustains effort, but it also suppresses the willingness to test demand honestly. The most passionate owners are often the ones who most need the evidence question asked.
- *“Get a loan and scale fast.”* Only when the cash cycle is short and margin is proven. Borrowing to buy demand that has not been demonstrated is the most common way small owners lose personal assets.
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4. Finalized Prototype Scope and Implementation Profile

The final prototype intentionally simplifies the user interaction while keeping a substantial Prolog knowledge base. A normal screening uses seven short screens. Those answers create working-memory facts that activate at least 25 expert rules. The system remains a first-pass screening tool: it evaluates the current form of a proposed small business and identifies what should be validated next; it does not predict business success.

Prototype element	Finalized design
User interaction	7 short screens in a normal evaluation
Fixed domain facts	25 explicit Prolog facts (minimum requirement exceeded)
Session/working facts	Approximately 12–18 facts created from one evaluation
Expert rules	25 IF–THEN rules
Certainty factors	Every reasoning/final rule carries a CF from –1.0 to +1.0
Inference	Prolog rule-based reasoning; backward chaining is the default query strategy
Primary outcomes	PROCEED; PROCEED WITH CAUTION; FURTHER VALIDATION REQUIRED; NOT RECOMMENDED
Negative subtype	NOT RECOMMENDED IN THIS FORM when the plan can be changed and reassessed
Explanation	Top three positive reasons, top three concerns, and an optional

	triggered-rule trace
Validation	Six representative expert test cases

4.1 Seven-screen interaction

Screen	Information collected	Facts created
1. Business profile	Business type; new business or branch; physical/online/hybrid.	business_type, venture_type, channel_type
2. Demand and price	Evidence of paying demand; target customer clarity; repeat demand; whether the price has been accepted/benchmarked.	demand_evidence, target_customer, repeat_demand, price_status
3. Owner readiness	Direct experience; available time; whether an experienced helper/partner is committed; reversibility of the start.	owner_experience, owner_time, experienced_support, reversibility
4. Money	Startup cost; monthly operating cost; capital available; funding source; gross-margin level.	startup_cost, monthly_cost, capital_available, funding_source, margin_level
5. Survival and cash	Expected break-even months; personal runway; customer payment/cash-cycle pattern.	breakeven_months, personal_runway, cash_cycle
6. Competition and operations	Competition strength; differentiation; major dependency; location/channel suitability.	competition, differentiation, dependency, channel_fit
7. Legal/licensing	Whether requirements are known, routine, unknown, or blocking.	legal_status

The interface may place several short controls on one screen. This keeps the demonstration brief without reducing the number of facts available to the inference engine. “I don’t know” is permitted for critical questions and is treated as an explicit value, never silently converted to a positive answer.

4.2 Derived values calculated by the system

$\text{capital_requirement} = \text{startup_cost} + (6 \times \text{monthly_operating_cost})$

$\text{funded_months} = \max(0, \text{capital_available} - \text{startup_cost}) \div \text{monthly_operating_cost}$

$\text{survival_gap} = \text{funded_months} - \text{breakeven_months}$

These values are computed from user inputs. They are not asked as separate questions.

5. Final Knowledge-Base Facts

The knowledge base contains at least 20 unconditional domain facts as requested in the project guidance. The finalized version contains 25 fixed facts before any user answers are added.

Session facts from the seven screens are additional working-memory facts.

`supported_business(service).`

`supported_business(retail).`

`supported_business(online).`

`supported_business(education).`

`supported_business(professional_service).`

`supported_business(food).`

`supported_business(owner_trade).`

`safe_funding(own_savings).`

`safe_funding(family_interest_free).`

`safe_funding(formal_affordable).`

`dangerous_funding(informal_high_interest).`

`dangerous_funding(pawned_asset).`

`dangerous_funding(home_secured_loan).`

`demand_level(none).`

`demand_level(anecdotal).`

`demand_level(observed).`

`demand_level(validated).`

cash_cycle_type(advance_payment).

cash_cycle_type(on_delivery).

cash_cycle_type(long_credit).

cash_cycle_type(heavy_stock).

legal_value(none_required).

legal_value(routine).

legal_value(unknown).

legal_value(blocking).

5.1 Example session facts

business_type(food).

demand_evidence(validated).

owner_experience(direct).

capital_available(2500000).

funding_source(own_savings).

competition(many_strong).

differentiation(strong).

legal_status(routine).

6. Intermediate Conclusions

Conclusion	Possible values	Basis
Market attractiveness	high moderate low unknown	Primarily demand evidence, repeat demand, target clarity and price validation.
Competitive position	strong adequate weak unknown	Competition and a specific, defensible differentiation.
Financial readiness	ready marginal not_ready unknown	Capital coverage, funding source, margin and cash cycle.

Owner readiness	strong adequate weak unknown	Direct experience, time availability and reversibility/experienced support.
Operational readiness	ready marginal not_ready unknown	Channel/location fit, dependency and ability to operate when customers need the business.
Overall risk	low moderate high critical	Survival gap, household runway and concentration/dependency risk.
Validation gap	list of unknown/assumed critical items	Demand, price, capital requirement and legal/licensing information that still needs checking.

7. Final Rule Knowledge Base (25 Rules)

Each rule uses the lecture-style form IF <evidence> THEN <hypothesis> {CF}. The CF expresses the expert's belief in the rule conclusion when its evidence is present; it is not a probability of business success.

R01 — IF demand_evidence = validated AND repeat_demand = recurring THEN market_attractiveness = high CF +0.8. Strongest practical evidence of a repeatable market.

R02 — IF demand_evidence $\in$ {observed, validated} AND target_customer = specific THEN market_attractiveness = moderate/high CF +0.6. Real evidence plus a reachable customer group supports the market conclusion.

R03 — IF demand_evidence $\in$ {none, anecdotal} THEN market_attractiveness = low; validation_gap += test_demand CF -0.8. Interest or encouragement is not willingness to pay.

R04 — IF price_status = assumed THEN validation_gap += validate_price CF -0.4. Financial conclusions are weak until the selling price is checked.

R05 — IF competition = many_strong AND differentiation $\in$ {none, price_only} THEN competitive_position = weak CF -0.8. A small entrant cannot rely on an undifferentiated price fight.

R06 — IF competition $\in$ {few, many_weak} AND differentiation = strong THEN competitive_position = strong CF +0.6. A specific reason to choose the business can offset crowding.

R07 — IF capital_available $\geq$ capital_requirement AND safe_funding(funding_source) THEN financial_readiness = ready CF +0.8. The plan is funded through startup and six months of operation from a safe source.

R08 — IF capital_available $\geq$ 50% of capital_requirement AND capital_available < capital_requirement THEN financial_readiness = marginal/not_ready CF -0.6. The plan is underfunded and should be reduced or delayed.

R09 — IF capital_available < 50% of capital_requirement THEN overall_risk = high; red_flag = severe_underfunding CF -0.8. The plan is unlikely to stay open long enough to test the market.

R10 — IF dangerous_funding(funding_source) THEN recommendation = NOT RECOMMENDED CF -1.0. Absolute override: risky funding can endanger the household.

R11 — IF margin_rate < effective_funding_cost THEN recommendation = NOT RECOMMENDED CF -1.0. Absolute override: each sale destroys value at the financing cost.

R12 — IF margin_level $\in$ {adequate,strong} AND cash_cycle $\in$ {advance_payment,on_delivery} THEN financial_readiness = ready CF +0.6. Healthy margin and early cash collection make mistakes more survivable.

R13 — IF cash_cycle $\in$ {long_credit,heavy_stock} AND financial_readiness $\neq$ ready THEN overall_risk = high CF -0.6. Money is tied up before cash returns to the business.

R14 — IF breakeven_months > funded_months THEN overall_risk = high; survival_runway = insufficient CF -0.8. The business is expected to run out of funded time before break-even.

R15 — IF owner_experience = direct THEN owner_readiness = strong CF +0.8. Hands-on experience reduces operating and quality-control errors.

R16 — IF owner_experience = none AND reversibility = low THEN owner_readiness = weak CF -0.8. Learning is expensive when the commitment cannot be reversed.

R17 — IF owner_experience = none AND reversibility = high THEN owner_readiness = adequate CF +0.4. Low-cost reversible starts allow the owner to learn safely.

R18 — IF owner_time = insufficient AND channel_type = walk_in THEN operational_readiness = not_ready CF -0.6. A walk-in business must be available during customer hours.

R19 — IF dependency = high THEN overall_risk = high; red_flag = concentration CF -0.6. One customer, supplier, person or platform can become a single point of failure.

R20 — IF personal_runway < 3 AND breakeven_months > 6 THEN overall_risk = high; red_flag = household_exposure CF -0.8. Household pressure can force damaging short-term business decisions.

R21 — IF legal_status = blocking THEN recommendation = NOT RECOMMENDED CF -1.0. Absolute override: the business cannot legally operate in the proposed form.

R22 — IF critical information is unknown/assumed (demand, price, capital requirement or legal status) THEN recommendation = FURTHER VALIDATION REQUIRED CF 0.0. The system should report insufficient basis rather than guess.

R23 — IF market is positive AND finance ready AND owner adequate/strong AND operations acceptable AND no override/red flag THEN recommendation = PROCEED CF +0.8. Nothing in the screening currently argues against continuing to detailed planning.

R24 — IF core factors are positive AND at most two weaknesses are manageable AND no absolute override THEN recommendation = PROCEED WITH CAUTION CF +0.6. The opportunity may continue only with stated mitigations.

R25 — IF two or more severe unmitigated risks are present AND no absolute override already fired THEN recommendation = NOT RECOMMENDED IN THIS FORM CF -0.8. The current scale/model should change before reassessment.

7.1 Prolog representation example

rule(r01, market_attractiveness(high), 0.8) :-

 demand_evidence(validated),

 repeat_demand(recurring).

rule(r21, recommendation(not_recommended), -1.0) :-

 legal_status(blocking).

8. Certainty Factor Model

The prototype follows the certainty-factor representation taught in the Expert Systems lecture: a CF is a numerical measure of the expert's belief, ranging from -1.0 (definitely false) to +1.0 (definitely true). Rules are represented as IF <evidence> THEN <hypothesis> {cf}.

CF	Interpretation
-1.0	Definitely not / absolute negative
-0.8	Almost certainly not
-0.6	Probably not

−0.4	Maybe not
−0.2 to +0.2	Unknown / insufficient basis
+0.4	Maybe
+0.6	Probably
+0.8	Almost certainly
+1.0	Definitely

8.1 Simplified CF handling for the prototype

Every expert rule stores one CF alongside its conclusion.

CF values describe confidence in the rule conclusion, not probability of business success.

The prototype does not use a complex accumulated-CF scoring formula. Intermediate conclusions are produced by the applicable expert rules, and the final recommendation rule has its own CF.

If several rules explain the same conclusion, the explanation shows the strongest applicable rule first and lists the others as supporting or conflicting evidence.

Absolute overrides (R10, R11 and R21) bypass all positive evidence.

Unknown critical information triggers R22 and blocks PROCEED until the missing information is validated.

The result screen shows the plain-language confidence label first, followed by the numeric CF in brackets.

9. Final Recommendations and Decision Logic

Recommendation	Meaning and certainty
PROCEED	Almost certainly (+0.8) — Demand and the critical readiness dimensions are sufficiently supported; no override or major red flag is present. This means continue to detailed planning, not that success is guaranteed.

PROCEED WITH CAUTION	Probably (+0.6) — The core case is positive but one or two manageable weaknesses require explicit mitigation before or during launch.
FURTHER VALIDATION REQUIRED	Unknown/insufficient basis (0.0) — A critical input such as demand, price, true capital requirement or legal status is unknown/assumed. The system identifies exactly what to test next.
NOT RECOMMENDED	Definitely/strongly negative (−1.0 where an absolute override applies) — A blocking legal condition, dangerous funding structure or destructive margin condition makes the current decision unacceptable.
NOT RECOMMENDED IN THIS FORM	Almost certainly not (−0.8) — Two or more severe but potentially correctable risks exist. The plan should be changed (scale, model, dependency, timing) and then reassessed.

9.1 Priority order

Check absolute overrides: dangerous funding, destructive margin, blocking legal status.

Check critical unknowns: if demand/price/capital/legal status is unknown, return FURTHER VALIDATION REQUIRED.

Infer the intermediate conclusions using the applicable rules.

Apply the positive final rule (PROCEED) or caution rule if the evidence supports them.

If multiple severe unmitigated risks remain, return NOT RECOMMENDED IN THIS FORM.

10. Explanation Module Requirements

Overall recommendation and CF label/number.

The three strongest positive reasons.

The three strongest concerns or red flags.

Any missing/unknown critical information.

A specific next action for every concern.

Visible arithmetic for capital requirement and funded months when money inputs are available.

A “Show reasoning” control that lists the triggered rules (for example: R01 → market high (+0.8)).

The main result screen should use plain business language. Technical terms such as “CF” and “rule trace” may appear in the optional reasoning view for the expert/examiner.

11. User Interface and Interaction Requirements

A normal evaluation is presented as seven short screens with a visible progress indicator (1/7, 2/7, ..., 7/7). Most answers are dropdowns, radio buttons, yes/no choices or small numeric fields. A lecturer should be able to complete a representative demonstration in about one to two minutes; a normal user should typically finish within five minutes.

The user can go back and change an answer before or after seeing the result.

Critical questions include an “I don’t know” option.

Irrelevant controls are hidden where practical (for example, walk-in location logic is not applied to a purely online business).

Money and time are entered numerically; categories are not used where the rule requires arithmetic.

The result can be recalculated immediately after one answer is changed.

The interface is mobile-friendly and uses simple English in the prototype.

12. Expert Validation Scenarios

Case 1 — Home-based tuition — Validated demand; teacher has direct experience; fees in advance; very low fixed cost; safe funding; no licensing blocker. Expected: PROCEED CF +0.8. Demonstrates strong demand, owner experience and self-funding.

Case 2 — Bubble tea outlet — No demand test; no direct experience; strong competitors; irreversible three-year lease; capital below true requirement. Expected: NOT RECOMMENDED IN THIS FORM CF -0.8. Multiple severe but correctable plan-design risks; the scale/lease should change before reassessment.

Case 3 — Cloud kitchen — Experienced owner and observed office interest, but capital is under half of the six-month requirement, one-platform dependency, and food registration is not yet confirmed. Expected: FURTHER VALIDATION REQUIRED CF 0.0. Legal status and funding structure must be clarified before a positive recommendation.

Case 4 — Home-based web agency — Very low downside, but no clients, no price validation and no proven customer-acquisition route. Expected: FURTHER VALIDATION REQUIRED CF 0.0. The system should request paid demand/price validation rather than guess.

Case 5 — Grocery shop with pawned-jewellery funding — Pawned-asset funding at high cost; expected margin below funding cost; owner absent during trading hours. Expected: NOT RECOMMENDED CF -1.0. R10/R11 absolute override.

Case 6 — Second salon branch — Proven model, experienced owner, safe retained-profit funding and good cash collection; one uncontracted senior stylist is a major dependency. Expected: PROCEED WITH CAUTION CF +0.6. The dependency is serious but directly mitigable with a written commitment and backup staff plan.

13. Final Functional Requirements

ID	Requirement
FR-01	Start a new screening and identify whether the case is a new business or an additional branch.
FR-02	Collect the seven-screen set of business, demand, owner, financial, survival, competition/operations and legal inputs.
FR-03	Allow “I don’t know” on critical non-numeric questions and record it as an explicit fact.
FR-04	Convert user answers into session facts and calculate capital_requirement, funded_months and survival_gap.
FR-05	Apply the 25 expert IF–THEN rules and their certainty factors to derive intermediate conclusions and a final recommendation.
FR-06	Support the four primary recommendation categories and the “NOT RECOMMENDED IN THIS FORM” subtype.
FR-07	Display a plain-language certainty label plus the numeric CF for the final recommendation.
FR-08	Display the three strongest positive reasons, three strongest concerns, missing information and recommended next actions.
FR-09	Provide a “Show reasoning” view containing triggered rule IDs, conclusions and CF values.
FR-10	Allow the user to change an answer and immediately rerun the assessment.

FR-11	Keep the knowledge base (facts/rules/CFs) separate from the interface code so it can be edited without rewriting the UI.
FR-12	Allow the user to reset and start a new assessment; a printable/savable summary may be included if time permits.

14. Non-Functional Requirements

Requirement	Specification
Usability	Seven screens; normal completion target under five minutes; short demo target about one to two minutes.
Performance	Result generated locally/in-process within approximately one second after the final answer.
Transparency	Every displayed conclusion must be traceable to one or more triggered rules and user facts.
Maintainability	Facts, rules and CF values are stored separately from interface code.
Platform	Mobile-friendly web interface for the prototype.
Privacy	Do not require names, NIC numbers, addresses, phone numbers or named third parties.
Robustness	Missing required numeric input must produce a visible validation message; the system must not invent defaults.
Language	Prototype in clear English. Sinhala/Tamil question text can be a future extension without changing the rule logic.

15. Boundaries, Disclaimer, and Responsible Use

B0SS provides a preliminary, rule-based screening of a proposed small-business opportunity. It applies decision rules captured from a business advisor to the information supplied by the user. It does not analyse current market data, does not predict or guarantee business success or failure, and does not provide legal, tax, accounting, investment or lending advice. A PROCEED result means only that the screening rules do not currently identify a reason to stop further investigation. Before an irreversible commitment, the user should verify demand, figures and licensing with appropriate real-world evidence and qualified professionals where necessary.

Never output a percentage probability of success.

Never recommend taking a loan, pledging a home/asset, leaving employment, choosing a legal structure, hiring/firing a named person, or signing a lease.

Never convert unknown demand or unknown legal status into a positive answer.

Do not collect unnecessary identifying information about the user, customers, suppliers or employees.

16. Knowledge Validation and Sign-Off

After implementation, the expert should review the exact Prolog facts, all 25 rules, every CF value, the six validation cases, and the explanation wording. A rule is considered validated when the expert confirms that the system reaches the expected conclusion for the representative cases and can explain which facts/rules produced it.

At least 20 knowledge-base facts present (finalized version: 25 fixed facts).

At least 20 expert rules present (finalized version: 25 rules).

CF range and labels aligned with the lecture model (-1.0 to $+1.0$).

Absolute overrides tested.

Unknown-information path tested.

All six validation cases tested after implementation.

Explanation trace checked against the actual rules that fired.

Expert name: «expert name»

Signature/confirmation: _____

Date: _____

Knowledge engineer/interviewer: «your name»

Date: _____

Appendix C — Updated Interview / Design Record

Session 1 date/time: «date» — Expert profile, scope/exclusions, natural decision process, critical factors and red flags were captured.

Session 2 date/time: «date» — The knowledge base was simplified to 25 fixed facts and 25 expert rules. Certainty factors were aligned with the lecture representation: IF evidence THEN hypothesis {cf}, using values from -1.0 to $+1.0$.

Session 3 / validation date-time: «date» — The user interaction was reduced to seven short screens; explanation output, four primary recommendation categories, the “NOT RECOMMENDED IN THIS FORM” subtype, and six validation cases were finalized.

Remaining implementation validation: run all six cases against the completed Prolog knowledge base and confirm the triggered-rule trace with the expert.